

Competency

Allocated teaching periods	22
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- To find the area of plane surface and surface area and volume of solids then solving real life problems based on cost estimation.

Learning Outcomes

- To estimate the cost of carpeting, constructing and gravelling the paths, painting, papering, plastering etc. related to real life situations
- To find the problems based on area of walls, floor and ceiling of the room
- To find the cross sectional area, LSA , TSA and volume of prisms and solve the related problems
- To prepare reports and project works on problems of area related to daily life situations

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To define the area of plane surface - To tell the formula of finding the area of cross, inner and outer paths - To relate the area, rate of cost per sq. units and the total cost of working (carpeting, plastering etc.) - To tell the formula of finding the area of four walls of room - To define cross-sectional area of prism - To recall the general formulae of finding the LSA, TSA and volume of the prism
2.	Understanding (U)	- To area of plane figures (rectangle, square, parallelogram, triangle, quadrilateral, trapezium and circle) - To calculate the area of paths (cross, inner and outer paths) - To find the area of floor, walls and ceiling of room - To find the volume of prism - To find the surface area of prisms
3.	Application (A)	- To find the area of paths and cost of graveling, covering with stones - To find the area of floor and cost of carpeting - To find the area of walls and estimate the cost of painting, papering, carpeting - To solve the problems related to the volume and surface area of prisms - To estimate the number of bricks and cost required to construct the walls etc.
4.	High Ability (HA)	- To mathematize the daily life problems related to area and volume then solve them. - To prepare the projects works and reports and present in the class

Required Teaching Materials/ Resources

Colourful chart-paper with definitions and formulae, scale, scissors, pencil, geo-board, tangram, graph paper, models of various prisms, box, cartoon, measuring tape, ICT tools etc.

Pre-knowledge: Perimeter and area of rectangle, parallelogram, volume of cube, cuboid etc.

A. Area of plane surfaces and paths

Teaching Activities

1. Divide the students into 4/5 groups and give them the figure of rectangle, square, parallelogram, rhombus, circle, quadrilateral, trapezium etc. on the sheet of paper and tell them to recall the perimeter and area formula of related figures and call for presentation.

2. Discuss upon the following formulae with related figures and models.

- (i) Area of rectangle = length (l) $\times$ breadth (b)
- (ii) Area of square = (side)² or $\frac{1}{2}$ (diagonal)²
- (iii) Area of parallelogram = base (b) $\times$ height (h)
- (iv) Area of triangle = $\frac{1}{2} \times$ base (b) $\times$ height (h)
- (v) Area of right angled triangle = $\frac{1}{2} \times$ base (b) $\times$ perpendicular (p)
- (vi) Area of equilateral triangle = $\frac{\sqrt{3}}{4} a^2$ where 'a' is the side length
- (vii) Area of triangle = $\sqrt{s(s-a)(s-b)(s-c)}$ where a, b and c are the lengths of sides and s is the semi-perimeter.
- (viii) Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$ where d_1 and d_2 are the diagonals of the rhombus.
- (ix) Area of kite = $\frac{1}{2} \times d_1 \times d_2$ where d_1 and d_2 are the diagonals of the kite.
- (x) Area of trapezium = $\frac{1}{2} \times h (a + b)$ where 'a' and 'b' are the lengths of parallel sides and h is the height of the trapezium.
- (xi) Area of quadrilateral = $\frac{1}{2} \times d (h_1 + h_2)$ 'h₁' and 'h₂' are the heights of two triangles on the same base which the diagonal (d) of quadrilateral
- (xii) Area of circle = $\frac{1}{2} \pi d^2$ or πr^2
- (xiii) Area of semi-circle = $\frac{1}{8} \pi d^2$ or πr^2

3. With discussion, derive the following formula of finding the area of pathways

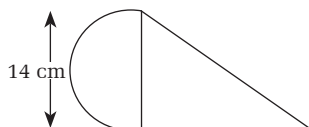
- (i) Area of path running outside the rectangle = $2d (l + b + 2d)$
- (ii) Area of path running inside the rectangle = $2d (l + b - 2d)$
- (iii) Area of crossing paths = $d (l + b - d)$
- (iv) Area of path running outside the circle = $\pi d (2r + d)$ or $\pi (R^2 - r^2)$
- (v) Area of path running inside the circle = $\pi d (2r - d)$

4. With examples, discussion upon the following formulae

- (i) Total cost (T) = Length (l) $\times$ Rate (R) per unit
- (ii) Total cost (T) = Area (A) $\times$ Rate (R) per square unit
- (iii) Total cost (T) = Number of bricks/stone (N) $\times$ Rate (R) per brick/stone
- (iv) Number of bricks/stone (N) =
$$\frac{\text{Area of path (A)}}{\text{Base area of each brick/stone (a)}}$$

Solution of selected problems from Vedanta Excel in Mathematics

1. Calculate the area of shaded region.



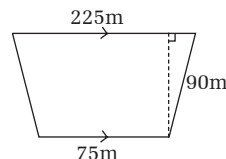
Solution:

$$\text{Area of triangular region } (A_1) = \frac{1}{2} \times b \times p = \frac{1}{2} \times 10 \text{ cm} \times 14 \text{ cm} = 70 \text{ cm}^2$$

$$\text{Area of semi-circular region } (A_2) = \frac{1}{8} \pi d^2 = \frac{1}{8} \times \frac{2}{7} \times (14 \text{ cm})^2 = 77 \text{ cm}^2$$

$$\therefore \text{Area of the shaded region } (A) = A_1 + A_2 = 70 \text{ cm}^2 + 77 \text{ cm}^2 = 147 \text{ cm}^2$$

2. The adjoining figure is a park in the shape of a trapezium. Calculate the cost of paving the park with stones at Rs 80 per sq.m.

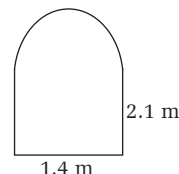


Solution:

$$\text{Area of the park } (A) = \frac{1}{2} \times h(a + b) = \frac{1}{2} \times 90 \text{ m} (225 + 75) \text{ m} = 13,500 \text{ m}^2$$

$$\text{Cost of paving the park with stones } (T) = \text{Area } (A) \times \text{Rate } (R) = 13,500 \times \text{Rs } 80 = \text{Rs } 10,80,000$$

3. The frame of the given window is made up of a rectangle with a semi-circular top. Find its perimeter and area.



Solution:

$$(i) \text{ Perimeter of lowermost rectangular shape } (P_1) = l + 2b = 1.4 \text{ m} + 2 \times 2.1 \text{ m} = 5.6 \text{ m}$$

$$\text{Perimeter of uppermost semi-circular top } (P_2) = \frac{1}{4} \times 2\pi r^2 = \frac{1}{2} \times \frac{22}{7} \times 0.7 \text{ m} = 2.2 \text{ m}$$

$$\therefore \text{The perimeter of the frame } (P) = P_1 + P_2 = 5.6 \text{ m} + 2.2 \text{ m} = 7.8 \text{ m}$$

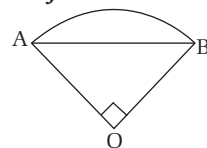
$$\text{Area of the frame } (A) = l \times b = 1.4 \text{ m} \times 2.1 \text{ m} = 2.94 \text{ m}^2$$

4. The shape alongside is a one-quarter of a circle with radius of 14 cm. find

(i) the length of the arc AB. (ii) the perimeter of the figure.

(iii) the area of the figure. (iv) the area of $\triangle AOB$.

(v) the area of the shaded segment.



Solution:

$$(i) \text{ The length of the arc AB } = \frac{1}{4} \times \text{circumference of circle with radius OA} \\ = \frac{1}{4} \times 2\pi r = \frac{1}{2} \times \frac{22}{7} \times (14 \text{ cm}) = 22 \text{ cm}$$

$$(ii) \text{ The perimeter of the figure } (P) = \text{length of arc AB} + 2r = 22 \text{ cm} + 2 \times 14 \text{ cm} = 50 \text{ cm}$$

$$(iii) \text{ The area of the figure } = \frac{1}{4} \times \pi r^2 = \frac{1}{4} \times \frac{22}{7} \times (14 \text{ cm})^2 = 154 \text{ cm}^2$$

$$(iv) \text{ The area of } \triangle AOB = \frac{1}{2} \times 14 \text{ cm} \times 14 \text{ cm} = 98 \text{ cm}^2$$

$$(v) \text{ The area of the shaded segment } = 154 \text{ cm}^2 - 98 \text{ cm}^2 = 56 \text{ cm}^2$$

5. A rectangular hall is 12 m long and 10 m broad. Find the length of carpet 2 m wide required for covering its floor. If the rate of carpet is Rs 110 per meter, find the cost of carpeting the floor.

Solution:

Here, length of hall (l) = 12 m and breadth of hall (b) = 10 m

∴ Area of the floor of the hall (A) = $l \times b = 12 \text{ m} \times 10 \text{ m} = 120 \text{ m}^2$

Also, width of carpet (b_1) = 2 m, length of carpet (l_1) = ?

We know, area of carpet required = area of floor of the hall

$$\text{or, } l_1 \times b_1 = 120 \text{ m}^2 \quad \text{or, } l_1 \times 2 \text{ m} = 120 \text{ m}^2 \quad \therefore b_1 = 60 \text{ m}$$

Again, rate of carpeting the floor (R) = Rs 110 per meter

∴ Total cost of carpeting the floor of the hall (T) = $l_1 \times R = 60 \times \text{Rs } 110 = \text{Rs } 6600$

- 6. A rectangular court is twice as long as its breadth and its perimeter is 540 m. Find the number of bricks of size 20 cm × 12 cm to pave the court. If the rate of cost of bricks is Rs 950 per 1000, find the cost of paving the court.**

Solution:

Let, the breadth of the court (b) be x m then length (l) = $2x$ m.

Now, perimeter (P) = $2(l + b)$ or, $540 \text{ m} = 2(2x + x)$ ∴ $x = 90 \text{ m}$ and $2x = 180 \text{ m}$ i.e., length (l) = 180 m and breadth (b) = 90 m

Also, area of the court (A) = $l \times b = 180 \text{ m} \times 90 \text{ m} = 16200 \text{ m}^2$

Base area of each brick (a) = $20 \text{ cm} \times 12 \text{ cm} = 0.2 \text{ m} \times 0.12 \text{ m} = 0.024 \text{ m}^2$

∴ No. of bricks required (N) = $\frac{\text{Area of path (A)}}{\text{Base area of each brick/stone (a)}} = \frac{16200 \text{ m}^2}{0.024 \text{ m}^2} = 675000$

Again, cost of 1000 bricks = Rs 950 or, cost of each brick = Rs 0.95

Hence, the cost of 675000 bricks = $675000 \times \text{Rs } 0.95 = \text{Rs } 6,41,250$.

- 7. A rectangular park is 250 m long and 140 m broad. A path 2 m wide is running around inside the park.**

(i) Calculate the cost of paving the path with stones at Rs 45 per sq. metre.

(ii) Calculate the cost of covering the empty space with turfs at Rs 25 per sq. metre.

Solution:

Here, length of the park (l) = 250 m,

breadth of the park (b) = 140 m

Width of the path running around inside the park (d) = 2 m

Now,

$$\begin{aligned} \text{(i) Area of the inner path (A)} &= 2d(l + b - 2d) \\ &= 2 \times 2 \text{ m}(250 \text{ m} + 140 \text{ m} - 2 \times 2 \text{ m}) = 1544 \text{ m}^2 \end{aligned}$$

Rate of cost of paving the path (R) = Rs 45 per sq. m

$$\begin{aligned} \therefore \text{Total cost of paving the park with stones (T)} &= \text{Area (A)} \times \text{Rate (R)} = 1544 \times \text{Rs } 45 \\ &= \text{Rs } 69,480 \end{aligned}$$

Hence, the total cost of paving the park with stones is Rs 69,480.

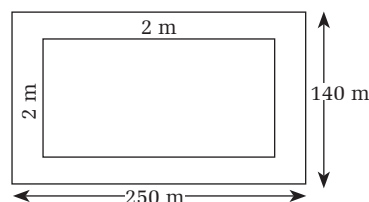
(ii) Area of entire park = $250 \text{ m} \times 140 \text{ m} = 35000 \text{ m}^2$

$$\begin{aligned} \therefore \text{Area of empty space} &= \text{area of the park} - \text{area of the path} \\ &= 35000 \text{ m}^2 - 1544 \text{ m}^2 = 33,456 \text{ m}^2 \end{aligned}$$

Again, rate of covering the empty space with turfs (R) = Rs 25 per sq. m

$$\begin{aligned} \therefore \text{Total cost of covering empty space with turfs (T)} &= \text{Area (A)} \times \text{Rate (R)} \\ &= 33,456 \times \text{Rs } 25 = \text{Rs } 8,36,400 \end{aligned}$$

Hence, the total cost of covering empty space with turfs is Rs 8,36,400.



8. The length of the side of a square land is 70 feet. For the purpose of the real-estate business, a path 10 feet wide inside the boundary of the land is made.

(i) Find the area of the path.

(ii) How many stones each of 2 feet long and 1.5 feet breadth are required to pave the path?

(iii) If a stone costs Rs 105, calculate the cost of paving the path.

Solution:

Here,

Length of the square land park (l) = 70 feet

Width of the path running around inside the boundary (d) = 10 feet

Now,

$$(i) \text{ Area of the inner path (A) } = 2d(l + b - 2d) \\ = 2 \times 10 \text{ feet } (70 \text{ feet} + 70 \text{ feet} - 2 \times 10 \text{ feet}) = 2400 \text{ feet}^2$$

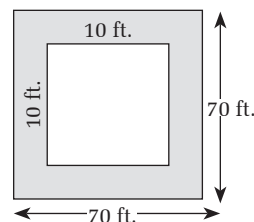
$$(ii) \text{ Area of each stone (a) } = 2 \text{ feet} \times 1.5 \text{ feet} = 3 \text{ feet}^2$$

$$\therefore \text{ No. of stones required } = \frac{\text{Area of path (A)}}{\text{Base area of each brick (a)}} = \frac{2400 \text{ feet}^2}{3 \text{ feet}^2} = 800$$

$$(iii) \text{ Cost of each stone (R) } = \text{Rs } 105$$

$$\therefore \text{ Total cost of paving the path (T) } = \text{No. of stones (N)} \times \text{Rate (R)} \\ = 800 \times \text{Rs } 105 = \text{Rs } 84,000$$

Hence, the total cost of paving the path with stones is RS 84,000.



9. The area of a square pond is 5625 m^2 and a 2 m wide path is made around the pond.

(i) Find the area of the path.

(ii) Calculate the number of tiles each of $40 \text{ cm} \times 20 \text{ cm}$ required to pave the path.

(iii) If the cost of a tile is Rs 35, find the cost of paving the path.

Solution:

Here,

$$(i) \text{ Area of the square pond (A) } = 5625 \text{ m}^2$$

$$\text{or, } l^2 = 5625 \text{ m}^2$$

$$\therefore l = 75 \text{ m}$$

$$(ii) \text{ Width of path around the pond (d) } = 2 \text{ m}$$

$$\therefore \text{ Area of the outer path (A) } = 2d(l + b + 2d) = 2 \times 2 \text{ m } (75 \text{ m} + 75 \text{ m} - 2 \times 2 \text{ m}) = 616 \text{ m}^2$$

$$(iii) \text{ Area of each tile (a) } = 40 \text{ cm} \times 20 \text{ cm} = 0.4 \text{ m} \times 0.2 \text{ m} = 0.08 \text{ m}^2$$

$$\therefore \text{ Required no. of tiles } = \frac{\text{Area of path (A)}}{\text{Base area of each brick (a)}} = \frac{616 \text{ m}^2}{0.08 \text{ m}^2} = 7700$$

$$(iv) \text{ Cost of each tile (R) } = \text{Rs } 35$$

$$\therefore \text{ Total cost of paving the path (T) } = \text{No. of tiles (N)} \times \text{Rate (R)} = 7700 \times \text{Rs } 35 = \text{Rs } 2,69,500$$

Hence, the total cost of paving the path with tiles is Rs 2,69,500.

10. The cost of construction of a path 5 m broad inside the boundary of a square lawn at Rs 36.25 per sq. metre is Rs 90,625. What is the cost of covering the empty space with turfs at the rate of Rs 20 per sq. metre?

Solution:

Here, width of the path (d) = 5 m

Rate of cost of construction of the path (R) = Rs 36.25

Total cost of construction of the path (T) = RS 90,625

Now, area of the path (A) = $\frac{T}{R} = \frac{\text{Rs } 90625}{\text{Rs } 36.25} = 2500 \text{ m}^2$

or, $4d(l - d) = 2500 \text{ m}^2$ or, $4 \times 5 \text{ m}(l - 5) = 2500 \text{ m}^2 \therefore l = 130 \text{ m}$

Again, length of empty space = $l - 2d = 130 \text{ m} - 2 \times 5 \text{ m} = 120 \text{ m}$

Area of empty space (A) = $(12 \text{ m})^2 = 144 \text{ m}^2$

Thus, the cost of covering the empty space with turfs = Area (A) $\times$ Rate (R)
= $14400 \times \text{Rs } 20 = \text{Rs } 2,88,000$

11. The given window frame is made up of iron plate of 7 cm wide. It is in the shape of a rectangle with a semi-circular top. Find the cost of painting the frame at 60 paisa per sq. cm.

Solution:

Here,

In the semi-circular top of the frame;

External diameter = 1.4 m = 140 cm

$\therefore$ External radius (R) = 70 cm

Internal diameter = 1.4 m - 2d = 140 cm - 14 cm = 126 cm

$\therefore$ Internal radius (r) = 63 cm

Now, area of semi-circular part of frame = $\frac{1}{2}\pi(R^2 - r^2) = \frac{1}{2} \times \frac{22}{7} (70^2 - 63^2) = 1463 \text{ cm}^2$

In the rectangular part of the frame;

Length of external rectangle (L) = 1.4 m = 140 cm and breadth (B) = 2.1 m - R
= 210 cm - 70 cm = 140 cm

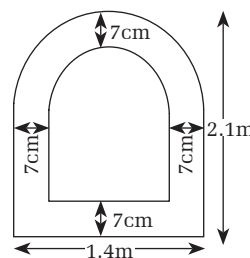
Length of internal rectangle (l) = 1.4 m - 2d = 126 cm and breadth (b) = B - d
= 140 cm - 7 cm = 133 cm

Area of rectangular frame = $L \times B - l \times b = 140 \text{ cm} \times 140 \text{ cm} - 126 \text{ cm} \times 133 \text{ cm} = 2842 \text{ cm}^2$

Area of the frame (A) = $1463 \text{ cm}^2 + 2842 \text{ cm}^2 = 4305 \text{ cm}^2$

Again, rate of painting (R) = 60 paisa per sq. cm = Rs 0.60 per sq. cm

$\therefore$ Total cost of painting the frame (T) = Area $\times$ Rate = $4305 \times \text{Rs } 0.60 = \text{Rs } 2583$



12. A wire in the form of rectangle 25.6 cm long and 18.4 cm wide is bent and reshaped into the form of circle. Calculate the change in the area in percent.

Solution:

Let the radius of the circle be r cm.

Then, perimeter of rectangular shape = circumference of circular shape

or, $2(l + b) = 2\pi r$

or, $2(25.6 \text{ cm} + 18.4 \text{ cm}) = 2 \times \frac{22}{7} r$

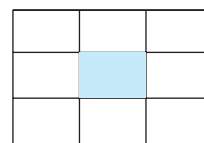
Now, area of rectangular shape = $l \times b = 25.6 \text{ cm} \times 18.4 \text{ cm} = 471.04 \text{ cm}^2$

Area of circular shape = $\frac{22}{7} \times (14 \text{ cm})^2 = 616 \text{ cm}^2$

$$\text{Change in area} = 616 \text{ cm}^2 - 471.04 \text{ cm}^2 = 144.96 \text{ cm}^2$$

$$\text{Area is increased by} = \frac{144.96 \text{ cm}^2}{471.04 \text{ cm}^2} \times 100\% = 30.8\%$$

- 13. The length of the adjoining rectangular park is two times its breadth. The cost of constructing two crossing paths running across the middle of the park at Rs 20 per sq. m is Rs 9000. If the cost of plastering the shaded portion of the park at Rs 35 per sq. m is Rs 315, find the cost of growing grasses in the empty spaces at Rs 12 per sq. metre.**



Solution:

Let the breadth of the park be x m then length $= 2x$ m.

Also, let the width of the crossing paths be d m.

Rate of constructing crossing paths (R) = Rs 20

Total cost of constructing crossing paths (T) = Rs 9000

Now, area of the crossing paths (A) =

$$\text{or, } d(l + b - d) = \quad \text{or, } d(2x + x - d) = 450 \therefore d(3x - d) = 450 \quad \dots (i)$$

Also, rate of plastering the square portion with side ' d m' (R) = Rs 35

Total cost of plastering (T) = Rs 315

$$\text{We have, area of the squared portion (A)} = \frac{T}{R} = \frac{\text{Rs } 315}{\text{Rs } 35}$$

$$\text{or, } d^2 = 9 \quad \therefore d = 3\text{m}$$

Putting the value of ' d ' in equation (i), we get

$$3(3x - 3) = 450 \quad \text{or, } x = 51 \therefore \text{breadth (b)} = 51 \text{ m and length (l)} = 2x = 102 \text{ m}$$

Again, area of empty space = area of whole park – area of crossing paths

$$= l \times b - d(l + b - d) = 102 \text{ m} \times 51 \text{ m} - 3\text{m}(102\text{m} + 51\text{m} - 3\text{m}) = 4752 \text{ m}^2$$

Thus, the total cost of growing grasses in the empty spaces = Area (A) $\times$ Rate (R)

$$= 4752 \times \text{Rs } 12 = \text{Rs } 57,024$$

Extra Questions

- A rectangular room is 8 m long and 5 m broad. Find the length of carpet 2.5 m wide required for carpeting the floor. If the rate of cost of carpet is Rs 375 per meter, find the cost of carpeting the floor. [Ans: 16m, Rs 6000]
- The length of a rectangular room is two times its breadth and its perimeter is 78 feet. Find the cost of carpeting its floor at Rs 25 per sq. feet. [Ans:Rs 8450]
- A rectangular garden 60 m long and 50 m broad is surrounded by 2.5 m wide path. Find the cost of paving the path at Rs 140 per sq. m [Ans: Rs 80,500]
- The cost of gravelling a path 3 m broad inside the boundary of a square park at Rs 35 per sq.m. is Rs 50,400. Find the cost of covering the empty space with turfs at Rs 25 per sq. m. [Ans:Rs 3,42,225]
- A rectangular garden is 62 m long and 48 m broad. Two paths each of 2 m wide, running across the middle of the garden, are at right angle. Calculate the cost of paving the paths by the bricks of size 18 cm by 12 cm at Rs 15 per brick. [Ans:Rs 1,50,000]

B. Area of 4 walls, floor and ceiling

Teaching Activities

1. Recall formula of finding the area of rectangle.
2. Tell the students to measure the internal and external lengths and breadths of the frame of white board and find the area of frame.
3. By taking classroom as an example, discuss about floor, ceiling and walls.
4. Make the group of students to measure the length, breadth and height of the room by measuring tape and tell them to find the area of floor, walls and ceiling.
5. Guide the students to discover the formula of area of 4 walls and ceiling of the room.
6. List the following formulae after discussion of the parts of the rooms with length = l , breadth = b and height = h :
 - (i) Area of floor = $l \times b$
 - (ii) Area of ceiling = $l \times b$
 - (iii) Area of four walls = $2lh + 2bh = 2h(l + b)$
 - (iv) Area of four walls = $2(l + b)h = \text{Perimeter of floor (P)} \times h$
 - (v) Area of 4 walls and ceiling = $2h(l + b) + lb$
 - (vi) Area of 4 walls, floor and ceiling = $2h(l + b) + lb + lb = 2h(l + b) + 2lb$

Solution of selected problems from Vedanta Excel in Mathematics

1. A rectangular room is 10 m long, 8 m wide and 5 m high. Find the cost of colouring its walls and ceiling at Rs 65 per sq. metre.

Solution:

Here, length of room (l) = 10 m, breadth (b) = 8 m and height (h) = 5 m

Now, area of walls and ceiling = $2h(l + b) + lb = 25m(10m + 8m) + 10m \times 8m = 260m^2$

Again, rate of colouring (R) = Rs 65 per sq. metre

$\therefore$ Total cost of colouring 4 walls and ceiling (T) = Area (A) $\times$ Rate (R) = $260 \times \text{Rs } 65 = \text{Rs } 16,900$

2. The cost of plastering the walls and ceiling of a room at Rs 20 per sq. feet is Rs 14,400. Find the cost of colouring the walls and the ceiling at Rs 16 per sq. feet.

Solution:

Here, rate of plastering the 4 walls and ceiling (R) = Rs 20 sq. feet

Total cost of plastering the 4 walls and ceiling (T) = Rs 14,400

$\therefore$ Area of 4 walls and ceiling (A) = $\frac{T}{R} = \frac{\text{Rs } 14,400}{\text{Rs } 20} = 720m^2$

Again,

Rate of colouring the 4 walls and ceiling (R) = Rs 16 sq. feet

Total cost of plastering the 4 walls and ceiling (T) = Area (A) $\times$ Rate (R) = $720 \times \text{Rs } 16$
= Rs 11,520

3. A rectangular room is 8 m long, 6 m broad and 4 m high. It contains 2 windows of size $2m \times 1.5m$ each and a door of size $1m \times 4m$, find the cost of painting its walls and ceiling at Rs 54 per sq. metre.

Solution:

Here, the length of room (l) = 8 m, the breadth (b) = 6 m and height (h) = 4m

The rate of painting the walls and ceiling (R) = Rs 54 per sq. metre

Now, the area of four walls = $2h(l + b) = 2 \times 4m(8m + 6m) = 112 \text{ m}^2$

The area of ceiling = $lb = 8 \text{ m} \times 6 \text{ m} = 48 \text{ m}^2$

The area of its 2 windows = $2(2 \text{ m} \times 1.5 \text{ m}) = 6 \text{ m}^2$

The area of its 1 door = $1 \text{ m} \times 4 \text{ m} = 4 \text{ m}^2$

$\therefore$ The area of 4 walls and ceiling excluding 2 windows and a door = $(112 + 48 - 6 - 4) \text{ m}^2$
 $= 150 \text{ m}^2$

Again, rate of painting its walls and ceiling (T) = Area (A) $\times$ Rate (R) = $150 \times \text{Rs } 54 = \text{Rs } 8,100$

- 4. A rectangular room is 10 m long and 8 m wide. It has 2 windows each of size 2 m 2 m and a door of size 1.5 m $\times$ 4 m. If the cost of plastering its walls at Rs 55 per sq. metre is Rs 9,130, find the height of the room.**

Solution:

Here, the length of room (l) = 10 m and the breadth (b) = 8 m

The rate of plastering the walls (R) = Rs 55 per sq. metre

Total cost of plastering the walls (T) = Rs 9,130

The area of its 2 windows = $2(2 \text{ m} \times 2 \text{ m}) = 8 \text{ m}^2$

The area of its 1 door = $1.5 \text{ m} \times 4 \text{ m} = 6 \text{ m}^2$

Now, the area of four walls excluding windows and door = $\frac{T}{R} = \frac{\text{Rs } 9,130}{\text{Rs } 55}$

or, $2h(l + b) - 8 \text{ m}^2 - 6 \text{ m}^2 = 166 \text{ m}^2$

or, $2h(10 + 8) - 14 = 166 \quad \therefore h = 5 \text{ m}$

Hence, the height of the room is 5 m.

- 5. The cost of carpeting a square room at Rs 110 per sq. metre is Rs 5,390. If the cost of plastering its walls at Rs 56 per sq. metre is Rs 7,840, find the height of the room.**

Solution:

The area of floor of the square room = $\frac{T}{R} = \frac{5,390}{110} \text{ m}^2$

or, $l^2 = 49 \text{ m}^2 \quad \therefore l = 7 \text{ m}$ and $b = 7 \text{ m}$

Again, area of 4 walls = $\frac{T}{R} = \frac{\text{Rs } 7,840}{\text{Rs } 56}$

or, $2h(7 + 7) = 140 \text{ m} \quad \therefore h = 5 \text{ m}$

Hence, the height of the room is 5 m.

- 6. A rectangular room is twice as long as it is broad and its height is 4.5 m. If the cost of papering its walls at Rs 40 per sq. metre is Rs 6,480, find the cost of paving on its floor at Rs 150 per sq. metre.**

Solution:

Let the breadth of the room (b) = x m then length (l) = 2x, height (h) = 4.5 m

Now, area of the walls (A) = $\frac{T}{R} = \frac{6,480}{40} \text{ m}^2$

or, $2h(l + b) = 162 \quad \text{or, } 2 \times 4.5(2x + x) = 162 \quad \text{or, } x = 6$

∴ Length of room (l) = $2x = 12$ m and breadth (b) = 6 m

Again, area of floor (A) = $l \times b = 12\text{m} \times 6\text{m} = 72\text{m}^2$

Hence, the total cost of paving the floor (T) = Area (A) $\times$ Rate (R) = $72 \times \text{Rs } 150 = \text{Rs } 10,800$

7. The length of a rectangular room is twice its breadth and thrice its height. If the cost of carpeting the floor at Rs 112 per sq. metre is Rs 12,600, find the cost of plastering its walls and ceiling at Rs 60 per sq. metre.

Solution:

Let the length of the room (l) = x m.

Then the breadth of the room (b) = $\frac{x}{2}$ m and height (h) = $\frac{x}{3}$ m

Now, area of the floor (A) = $\frac{T}{R} = \frac{12,600}{112}\text{m}^2$

or, $l \times b = 112.5\text{m}^2$ or, $x \times \frac{x}{2} = 112.5$ ∴ $x = 15$.

The length of the room (l) = 15 m,

The breadth (b) = $\frac{15}{2}\text{m} = 7.5\text{m}$ and the height (h) = $\frac{x}{3}\text{m} = \frac{15}{3}\text{m} = 5\text{m}$

Again, area of the walls and the ceiling = $2h(l + b) + l \times b$

$$= 2 \times 5\text{m}(15\text{m} + 7.5\text{m}) + 15\text{m} \times 7.5\text{m} = 337.5\text{m}^2$$

Hence, the total cost of plastering the walls and ceiling (T) = Area (A) $\times$ Rate (R)

$$= 337.5 \times \text{Rs } 60 = \text{Rs } 20,250$$

Extra Questions

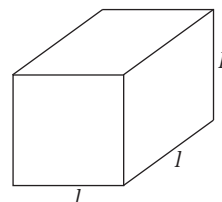
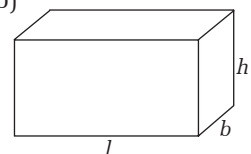
1. A rectangular room is 8 m long, 5 m broad and 4 m high. Find the cost of plastering its four walls at the rate of Rs 140 per sq. metre. [Ans: Rs 14,560]
2. A rectangular room is 15 m long, 10 m broad and 5 m high. It contains two windows of size 2 m $\times$ 1.5 m each and a door of size 1 m $\times$ 4 m, find the cost of painting its walls at Rs 40 per sq. metre [Ans: Rs 9,600]
3. A rectangular room is 10 m long and 8 m wide. It has 2 windows each of size 2 m $\times$ 2 m and a door of size 1.5 m $\times$ 4 m. If the cost of plastering its walls at Rs 55 per sq. metre is Rs 9,130, find the height of the room. [Ans: 5 m]
4. A house has three rooms. The length, breadth and height of the first room are 10 m, 9 m and 4.5 m, that of second rooms are 8 m, 6 m and 4.5 m and third room are 12 m, 7.5 m and 4.5 m. find the cost of painting the walls and ceilings of the rooms at the rate of Rs 50 sq. metre. [Ans: Rs 35,025]
5. The cost of carpeting a square room at the rate of Rs 75 per sq. metre is Rs 10,800. If the cost of plastering the walls at Rs 25 per sq. metre is Rs 6000, find the height of the room. [Ans: 5 m]
6. The length of a dining hall is twice its breadth and the breadth is twice its height. If the cost of carpeting the room at Rs 80 per sq. metre is Rs 10,240, what will be the cost of plastering its walls at Rs 40 per sq. metre? [Ans: Rs 8,640]

C. Area and volumes of solids

Teaching Activities

1. With solid shapes like cuboid and cube, ask the number of vertices, faces and edges.
2. Show the nets of solid object made by straws/ match sticks or through Geo-Gebra

3. Tell the students to draw the figure of cuboid and cube with corresponding formulae to find the volume and surface area on the chart paper
4. Discuss upon the volume and surface area of cuboid and cube as
 - (i) Base area of cuboid = $l \times b$
 - (ii) Lateral surface area of cuboid = area of 4 walls = $2h(l + b)$
 - (iii) Total surface area of cuboid = $2(lb + bh + lh)$
 - (iv) Total surface area of lidless cuboid = $2h(l + b) + lb$
 - (v) Total surface area of hollow cuboid = $2h(l + b)$
 - (vi) Volume of cuboid = $l \times b \times h$
 - (vii) Base area of cube = l^2
 - (viii) Lateral surface area of cube = area of 4 walls = $4l^2$
 - (ix) Total surface area of cube = $6l^2$
 - (x) Total surface area of lidless cube = $5l^2$
 - (xi) Total surface area of hollow cube = $4l^2$
 - (xii) Volume of cuboid = l^3



5. Give the solid prisms and their model/ nets to the students or show it through OHP using Geo-Gebra tool ask the definition and properties of prism
6. Discuss about its base, cross-section and lateral surface etc.
7. Conclude the following general formulae for prisms
 - (i) Volume of prism = Area of cross-section $\times$ height
 - (ii) Lateral surface (rectangular faces) area (L.S.A.) of prism = perimeter of cross section $\times$ height
 - (iii) Total surface area of prism = lateral surface area + $2 \times$ area of cross section

Solution of selected problems from Vedanta Excel in Mathematics

1. **A rectangular metallic block is 40 cm long, 24 cm broad and 10 cm high. How many pieces of rectangular slices each of 8 mm thick can be cast lengthwise from the block?**

Solution:

Here, length of block (l) = 40 m, breadth (b) = 24 m and height (h) = 10 m

Now, volume of block (V) = $l \times b \times h$ = 40 cm $\times$ 24 cm $\times$ 10 m = 9600 cm³

Again, volume of each slice (v) = 0.8 cm $\times$ 24 cm $\times$ 10 m = 192 cm³

$$\therefore \text{Number of slides (N)} = \frac{\text{Volume of block (V)}}{\text{Volume of each slice (v)}} = \frac{9600}{192} = 50$$

Hence, the required number of rectangular slices is 50.

2. **Calculate the cross sectional area, the lateral surface area, the total surface area and volume of the given prism.**

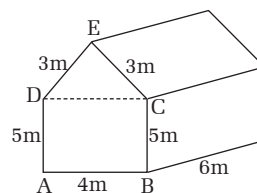
Solution:

Here

- (i) The area of cross section = area of rectangle ABCD + area of $\triangle EDC$

$$\text{The area of rectangle ABCD} = l \times b = 4\text{ m} \times 5\text{ m} = 20\text{ m}^2$$

$$\text{Semi-perimeter of } \triangle EDC = \frac{a + b + c}{2} = \frac{3\text{ m} + 3\text{ m} + 4\text{ m}}{2} = 5\text{ m}$$



- Area of $\Delta EDC = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{5(5-3)(5-3)(5-4)} = \sqrt{20}$
- $\therefore$ The area of cross section (A) = area of rectangle ABCD + area of ΔEDC
 $= 20 \text{ m}^2 + 4.47 \text{ m}^2 = 24.47 \text{ m}^2$
- (ii) Perimeter of the cross section (P) = $3\text{m} + 3\text{m} + 5\text{m} + 4\text{m} + 5\text{m} = 20\text{m}$
 Length of prism (h) = 6m
- $\therefore$ Lateral surface area (L.S.A.) = $P \times h = 20\text{m} \times 6\text{m} = 120\text{m}^2$
- (iii) Total surface area of prism = L.S.A. + $2A = 120\text{m}^2 + 2 \times 24.47\text{m}^2 = 168.94\text{m}^2$
- (iv) Volume of the prism (V) = $A \times h = 24.47\text{m}^2 \times 6\text{m} = 146.82\text{m}^3$
3. **A cubical water tank is filled in 1296 seconds at the rate of 1 litre per 6 seconds.**
 (i) **Calculate the internal volume and length of side of tank.**
 (ii) **Calculate the total internal surface area of the tank.**

Solution:

Here,

- (i) Amount of water filled in 6 seconds = 1 litre
 or, Amount of water filled in 1 second = $\frac{1}{6}$ litre
 $\therefore$ Amount of water filled in 1296 seconds = $\frac{1}{6} \times 1296$ litres = 216 litres
 We know that, 1 litre = 1000 cm^3
 $\therefore 216$ litres = $216 \times 1000\text{ cm}^3 = 216000\text{ cm}^3$
 Also,
 Internal volume of the tank = 216000 cm^3
 or, $l^3 = 216000\text{ cm}^3 \therefore l = 60\text{ cm}$
 So, the internal length of the tank is 60 cm .
- (ii) The total internal surface area of the tank = $6l^2 = 6 \times 60^2 = 21,600\text{ cm}^2$
 Hence, the total internal surface area of the tank is $21,600\text{ cm}^2$.
4. **A lidless rectangular water tank made of zinc plate is 2 m long, 1.5 m broad and 1 m high.**
 (i) **How many square metres of zinc plates are used in the tank?**
 (ii) **How many litres of water does it hold when it is full?**
 (iii) **Find the cost of zinc plates at Rs 1200 per sq.m.**

Solution:

Here, the length of tank (l) = 2 m , breadth (b) = 1.5 m and height (h) = 1 m

- (i) Surface area of lidless tank = $2h(l + b) + lb = 2 \times 1\text{m}(2\text{ m} + 1.5\text{ m}) + 2\text{ m} \times 1.5\text{ m} = 10\text{ m}^2$
 Thus, 10 sq. metre of zinc are used in the tank.
- (ii) Volume of the tank (V) = $l \times b \times h = 2\text{ m} \times 1.5\text{ m} \times 1\text{ m} = 3\text{ m}^3$
 $\therefore$ Capacity of the tank = 3×1000 litre = 3000 litres
- (iii) Total cost of zinc (T) = Area (A) $\times$ Rate (R) = $10 \times \text{Rs } 1200 = \text{Rs } 12,000$
 Hence, the cost of zinc plates is $\text{Rs } 12,000$.
5. **A rectangular carton is $80\text{ cm} \times 60\text{ cm} \times 40\text{ cm}$.**
 (i) **How many packets of soaps each of $10\text{ cm} \times 5\text{ cm} \times 4\text{ cm}$ can be kept inside the carton?**
 (ii) **By how many centimetres should the height of the carton be increased to keep 1200 packets of soaps?**

Solution:

- (i) Volume of carton (V) = $80 \text{ cm} \times 60 \text{ cm} \times 40 \text{ cm} = 1,92,000 \text{ cm}^3$
 Volume of each soap (v) = $10 \text{ cm} \times 5 \text{ cm} \times 4 \text{ cm} = 200 \text{ cm}^3$
 $\therefore$ No. of packets of soap that can be kept inside the carton = $\frac{V}{v} = \frac{1,92,000}{200} = 960$
- (ii) Let the height of carton to be increased be x cm.
 Then, height of the carton (h) = $(40 + x) \text{ cm}$
 Now, volume of carton = 1200 × volume of each soap
 or, $80 \text{ cm} \times 60 \text{ cm} \times (40 + x) \text{ cm} = 1200 \times 200 \text{ cm}^3$
 or, $x = 10$

Hence, the height of the carton should be increased by 10 cm to keep 1200 packets of soap.

6. **A rectangular metallic block is $50 \text{ cm} \times 20 \text{ cm} \times 8 \text{ cm}$. If it is melted and reformed in to a cubical block, find the length of edge of the cube.**

Solution:

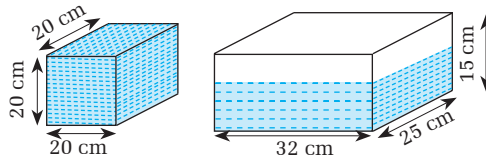
Let the edge of the cube be l cm.

Then, volume of cube = volume of the rectangular block

or, $l^3 = 50 \text{ cm} \times 20 \text{ cm} \times 8 \text{ cm} = 8000 \therefore l = 20 \text{ cm}$

Hence, the edge of the cube is 20 cm.

7. **In the given figure, a cubical vessel of length 20 cm is completely filled with water. If the water is poured in to a rectangular vessel of length 32 cm, breadth 25 cm and height 15 cm, find the height of water level in the rectangular vessel. How much more water is required to fill the rectangular vessel completely? ($1 \text{ l} = 1000 \text{ cm}^3$)**



Solution:

Volume of cubical vessel = $(20 \text{ cm})^3 = 8000 \text{ cm}^3$

Volume of water in rectangular vessel = $32 \text{ cm} \times 25 \text{ cm} \times h$

Now, $32 \text{ cm} \times 25 \text{ cm} \times h = 8000 \text{ cm}^3 \therefore h = 10 \text{ cm}$

Again, remaining height of rectangular vessel to be filled = $15 \text{ cm} - 10 \text{ cm} = 5 \text{ cm}$

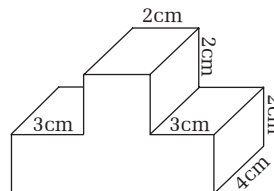
Then, volume of empty space of vessel = $32 \text{ cm} \times 25 \text{ cm} \times 5 \text{ cm} = 4000 \text{ cm}^3$

We know, $1000 \text{ cm}^3 = 1 \text{ l}$ $4000 \text{ cm}^3 = 4 \text{ l}$

Hence, the height of water level in the rectangular vessel is 10 cm and 4 l more water is required to fill the rectangular vessel completely.

Extra Questions

1. Calculate the total surface area and volume of the given prism. [Ans: 136 cm^2 , 80 cm^3]



2. A rectangular metallic block is 24 cm long, 20 cm broad and 15 cm high. How many pieces of rectangular slices each of 6 mm thick can be cast lengthwise from the block? [Ans: 40]
3. Three metallic cubes of edges 3 cm, 4 cm and 5 cm respectively are melted and reformed a bigger cube, what is the edge of bigger cube. [Ans: 6 cm]

4. Two cubes having volumes 125 cm^3 each are joined to form a cuboidal, find the total surface area of cuboidal. [Ans: 250 cm^2]
5. Vegetable ghee is stored in a rectangular vessel of internal dimensions $16 \text{ cm} \times 12 \text{ cm} \times 9 \text{ cm}$. It is transferred in to the identical cubical vessels. If the internal length of each cubical vessel is 6 cm , how many vessels are required to empty the rectangular vessel? [Ans: 8]

D. Estimation of number of cubes and cost required for building wall

Teaching Activities

1. Discuss upon the following formulae

(i) Volume of wall = $l \times b \times h$

(ii) Volume of each brick = $l_1 \times b_1 \times h_1$

$$\text{No. of bricks (N)} = \frac{\text{Volume of wall (V)}}{\text{Volume of each brick (v)}}$$

$$\text{Total cost of bricks (T)} = \text{No. of bricks (N)} \times \text{Rate (R) per brick}$$

$$\text{Space occupied by window} = \text{length} \times \text{height of window} \times \text{thickness of wall}$$

$$\text{Space occupied by door} = \text{length} \times \text{height of door} \times \text{thickness of wall}$$

Solution of selected problems from Vedanta Excel in Mathematics

1. *The dimensions of a part of compound wall are $30 \text{ m} \times 20 \text{ cm} \times 5 \text{ m}$. It contains 3 windows each of $2 \text{ m} \times 1.5 \text{ m}$. How many bricks of size $20 \text{ cm} \times 10 \text{ cm} \times 5 \text{ cm}$ are required to build the wall leaving 10% of the space for the cement work? Also, find the cost of bricks at the rate of Rs 18,000 per 1000 bricks.*

Solution:

Here, the length of wall (l) = $30 \text{ m} = 3000 \text{ cm}$, the breadth (b) = 20 cm and the height (h) = $5 \text{ m} = 500 \text{ cm}$

$$\text{Now, volume of wall (V)} = l \times b \times h = 3000 \text{ cm} \times 20 \text{ cm} \times 500 \text{ cm} = 3,00,00,000 \text{ cm}^3$$

Also, the length of window (l_1) = $2 \text{ m} = 200 \text{ cm}$,

the breadth (b_1) = width of wall = 20 cm and the height (h_1) = $1.5 \text{ m} = 150 \text{ cm}$

$$\therefore \text{Volume space occupied by of 3 windows} = 3(200 \text{ cm} \times 20 \text{ cm} \times 150 \text{ cm}) = 18,00,000 \text{ cm}^3$$

$$\text{Volume of wall excluding 3 windows} = 3,00,00,000 \text{ cm}^3 - 18,00,000 \text{ cm}^3 = 2,82,00,000 \text{ cm}^3$$

$$\text{Space of wall for cement work} = 10\% \text{ of } 2,82,00,000 \text{ cm}^3 = 28,20,000 \text{ cm}^3$$

$$\therefore \text{Volume of wall excluding the space occupied by windows and the space for cement work} = 2,82,00,000 \text{ cm}^3 - 28,20,000 \text{ cm}^3 = 2,53,80,000 \text{ cm}^3$$

$$\text{Again, volume of each brick (v)} = 20 \text{ cm} \times 10 \text{ cm} \times 5 \text{ cm} = 1000 \text{ cm}^3$$

$$\therefore \text{Number of bricks (N)} = \frac{\text{Volume of wall (V)}}{\text{Volume of each brick (v)}} = \frac{2,53,80,000}{1000} = 25,380$$

$$\text{Now, cost of 1000 bricks} = \text{Rs } 18,000 \quad \text{or, cost of 1 brick} = \text{Rs } 18$$

$$\therefore \text{The total cost of 25,380 bricks} = 25,380 \times \text{Rs } 18 = \text{Rs } 4,56,840$$

Hence, the required number of bricks is 25,380 and cost of bricks is Rs 4,56,840

2. *One of the four walls of a hall is 20 m long and 20 cm thick. If 4,700 bricks each of size $25 \text{ cm} \times 16 \text{ cm} \times 10 \text{ cm}$ are required to construct the wall containing two windows of size $2 \text{ m} \times 1.5 \text{ m}$, find the height of the wall.*

Solution:

Here

Here, the length of wall (l) = 20 m, the width (b) = 20 cm = 0.2 m

Now, volume of wall (V) = $l \times b \times h = 20 \text{ m} \times 0.2 \text{ m} \times h = 4h \text{ m}^3$

Also, the length of window (l_1) = 2 m, the height (h_1) = 1.5 m and the breadth (b_1) = width of wall = 20 cm = 0.2 m

$\therefore$ Volume space occupied by of 2 windows = $2(2 \text{ m} \times 0.2 \text{ m} \times 1.5 \text{ m}) = 1.2 \text{ m}^3$

Volume of wall excluding the windows = $(4h - 1.2) \text{ m}^3$

Again, volume of each bricks (v) = 25 cm $\times$ 16 cm $\times$ 10 cm

$$= 0.25 \text{ m} \times 0.16 \text{ m} \times 0.1 \text{ m} = 0.004 \text{ m}^3$$

$$\text{Again, number of bricks (N)} = \frac{\text{Volume of wall (V)}}{\text{Volume of each brick (v)}}$$

$$\text{or, } 4,700 = \frac{4h - 1.2}{0.004}$$

$$\text{or, } 18.8 = 4h - 1.2 \quad \therefore h = 5 \text{ m}$$

Hence, the height of the wall is 5 m.

3. A square room contains 288 m³ of air. The cost of carpeting the room at Rs 105 per sq. metre is Rs 6,720. Find the cost of painting its walls at Rs 45 per sq. metre.

Solution:

Let the length of room (l) = breadth of the room (b) = x m

Now, volume of the room (V) = 288 m³

$$\text{or, } l \times b \times h = 288 \text{ or, } x \cdot x \cdot h = 288 \quad \therefore x^2 h = 288 \quad \dots \text{equation (i)}$$

$$\text{Again, area of floor} = \frac{\text{Total cost of carpeting}}{\text{Rate of carpeting}} = \frac{6720}{105}$$

$$\text{or, } x^2 = 64 \quad \text{or, } x = 8 \quad \therefore l = b = 8 \text{ m}$$

Also, putting the value of x in equation (i), we get

$$8^2 h = 288 \quad \therefore h = 4.5 \text{ m}$$

$$\text{Again, the area of walls (A)} = 2h(l + b) = 2 \times 4.5(8 + 8) = 144 \text{ m}^2$$

$$\therefore \text{Total cost of painting the walls (T)} = \text{Area (A)} \times \text{Rate (R)} = 144 \times \text{Rs } 45 = \text{Rs } 6,480$$

Extra Questions

1. A square room contains 180 cu. metre of air. The cost of plastering its four walls at Rs 20 per sq. metre is Rs 2,400. Find the height of the room. [Ans: 5m]
2. After destruction by massive earthquake, the wall of length 40 m, height 5 m and width 20 cm was reconstructed. It contains two windows each of 2 m $\times$ 1.5 m and a gate of size 1.5 m $\times$ 4 m.
 - (i) Find the number of bricks each of 25 cm $\times$ 20 cm $\times$ 4 cm required to construct the wall.
 - (ii) Find the cost of the bricks at the rate of Rs 16,000 per 10000 bricks.
3. Mr. Gurung constructed a compound wall 40 m long and 20 cm wide by the bricks, each measuring 20 cm $\times$ 10 cm $\times$ 5 cm. If he paid Rs 4,32,000 at the rate of Rs 18,000 per 1000 bricks, find the height of the wall. [Ans: 3m]